

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE PHASE-I & II

TEST # 08

DATE : 30-09-2018

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

Corporate Office

ALLEN CAREER INSTITUTE

"SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-2757575 ☎ +91-744-2435003 ✉ info@allen.ac.in

Rajkot Centres

Kalawad Road

"ISHAVASYAM", B/h. Crystal Mall, Opp./O.C. Quarters,
Kalawad Road, Rajkot (Gujarat)-360005

☎ 0281-2562131 ☎ +91-85111-43783 ✉ rajkot@allen.ac.in

Limbudiwadi Centre

Opp. Landmark 2 Apartment, Limbudiwadi
Main Road, Off. Kalawad Road, Rajkot (Gujarat)- 360007

☎ 0281-2450228

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MINOR TEST # 08

Topics

DATE : 30 - 09 - 2018

PRE-MEDICAL NURTURE COURSE : PHASE - I & II – PATTERN : [NEET]

PHYSICS : Work, Power & Energy, Circular Motion

CHEMISTRY : Gaseous State, Chemical & Ionic Equilibrium

BIOLOGY : Families of Angiosperm, Anatomy of flowering Plants (up to vascular bundles),
 Digestion and Absorption

Next Up Coming Test

MINOR TEST # 09

Topics

DATE : 14 - 10 - 2018

PRE-MEDICAL NURTURE COURSE : PHASE - I & II – PATTERN : [NEET]

PHYSICS : Kinematics, NLM, Friction, Work Power & Energy

CHEMISTRY : Chemical Bonding, Gaseous State

BIOLOGY : Cell Biology and Plant Diversity (Upto Kingdom Protista) [Botany],
 Remaining Digestive System [Zoology]

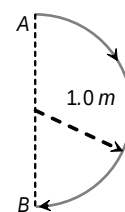
SPACE FOR ROUGH WORK

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BEWARE OF NEGATIVE MARKING

1. If the body is moving in a circle of radius r with a constant speed v , its angular velocity is
 (1) v^2/r (2) vr
 (3) v/r (4) r/v
2. Two racing cars of masses m_1 and m_2 are moving in circles of radii r_1 and r_2 respectively. Their speeds are such that each makes a complete circle in the same duration of time t . The ratio of the angular speed of the first to the second car is
 (1) $m_1 : m_2$
 (2) $r_1 : r_2$
 (3) $1 : 1$
 (4) $m_1 r_1 : m_2 r_2$
3. A body of mass m is moving in a circle of radius r with a constant speed v . The force on the body is $\frac{mv^2}{r}$ and is directed towards the centre.
 What is the work done by this force in moving the body over half the circumference of the circle
 (1) $\frac{mv^2}{r} \times \pi r$
 (2) Zero
 (3) $\frac{mv^2}{r^2}$
 (4) $\frac{\pi r^2}{mv^2}$
4. A stone of mass m is tied to a string of length l and rotated in a circle with a constant speed v . If the string is released, the stone flies
 (1) Radially outward
 (2) Radially inward
 (3) Tangentially outward
 (4) With an acceleration $\frac{mv^2}{l}$
5. A body is moving in a circular path with a constant speed. It has
 (1) A constant velocity
 (2) A constant acceleration
 (3) An acceleration of constant magnitude
 (4) An acceleration which varies with time
6. Two bodies of mass 10 kg and 5 kg moving in concentric orbits of radii R and r such that their periods are the same. Then the ratio between their centripetal acceleration is
 (1) R/r (2) r/R
 (3) R^2/r^2 (4) r^2/R^2
7. The ratio of angular speeds of minute hand and hour hand of a watch is
 (1) $1 : 12$ (2) $6 : 1$
 (3) $12 : 1$ (4) $1 : 6$
8. A particle is moving in a horizontal circle with constant speed. It has constant
 (1) Velocity
 (2) Acceleration
 (3) Kinetic energy
 (4) Displacement
9. An electric fan has blades of length 30 cm as measured from the axis of rotation. If the fan is rotating at 1200 r.p.m. The acceleration of a point on the tip of the blade is about
 (1) 1600 m/sec^2
 (2) 4740 m/sec^2
 (3) 2370 m/sec^2
 (4) 5055 m/sec^2
10. A particle of mass m is executing uniform circular motion on a path of radius r . If p is the magnitude of its linear momentum. The radial force acting on the particle is
 (1) $p m r$ (2) $\frac{r m}{p}$
 (3) $\frac{m p^2}{r}$ (4) $\frac{p^2}{r m}$

11. A mass of 2 kg is whirled in a horizontal circle by means of a string at an initial speed of 5 revolutions per minute. Keeping the radius constant the tension in the string is doubled. The new speed is nearly
 (1) 14 rpm
 (2) 10 rpm
 (3) 2.25 rpm
 (4) 7 rpm
12. A string breaks if its tension exceeds 10 newtons. A stone of mass 250 gm tied to this string of length 10 cm is rotated in a horizontal circle. The maximum angular velocity of rotation can be :-
 (1) 20 rad/s
 (2) 40 rad/s
 (3) 100 rad/s
 (4) 200 rad/s
13. A 500 kg car takes a round turn of radius 50 m with a velocity of 36 km/hr. The centripetal force is
 (1) 250 N
 (2) 750 N
 (3) 1000 N
 (4) 1200 N
14. A stone of mass of 16 kg is attached to a string 144 m long and is whirled in a horizontal circle. The maximum tension the string can withstand is 16 Newton. The maximum velocity of revolution that can be given to the stone without breaking it, will be
 (1) 20 ms^{-1}
 (2) 16 ms^{-1}
 (3) 14 ms^{-1}
 (4) 12 ms^{-1}
15. Find the maximum velocity for skidding for a car moved on a circular track of radius 100 m. The coefficient of friction between the road and tyre is 0.2
 (1) 0.14 m/s
 (2) 140 m/s
 (3) 1.4 km/s
 (4) 14 m/s
16. What is the value of linear velocity, if $\vec{\omega} = 3\hat{i} - 4\hat{j} + \hat{k}$ and $\vec{r} = 5\hat{i} - 6\hat{j} + 6\hat{k}$
 (1) $6\hat{i} + 2\hat{j} - 3\hat{k}$
 (2) $-18\hat{i} - 13\hat{j} + 2\hat{k}$
 (3) $4\hat{i} - 13\hat{j} + 6\hat{k}$
 (4) $6\hat{i} - 2\hat{j} + 8\hat{k}$
17. A particle is moving in a circle with uniform speed v . In moving from a point to another diametrically opposite point
 (1) The momentum changes by mv
 (2) The momentum changes by $2mv$
 (3) The kinetic energy changes by $(1/2)mv^2$
 (4) The kinetic energy changes by mv^2
18. A car of mass 800 kg moves on a circular track of radius 40 m. If the coefficient of friction is 0.5, then maximum velocity with which the car can move is
 (1) 7 m/s
 (2) 14 m/s
 (3) 8 m/s
 (4) 12 m/s
19. In 1.0 s, a particle goes from point A to point B, moving in a semicircle of radius 1.0 m (see figure). The magnitude of the average velocity is
 (1) 3.14 m/s
 (2) 2.0 m/s
 (3) 1.0 m/s
 (4) Zero
20. In a circus stuntman rides a motorbike in a circular track of radius R in the vertical plane. The minimum speed at highest point of track will be
 (1) $\sqrt{2gR}$
 (2) $2gR$
 (3) $\sqrt{3gR}$
 (4) $\sqrt{gR}$



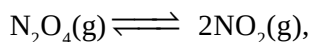
21. A car is moving with speed 30 m/sec on a circular path of radius 500 m. Its speed is increasing at the rate of 2m/sec^2 . What is the acceleration of the car
- (1) 2 m/sec^2
 (2) 2.7 m/sec^2
 (3) 1.8 m/sec^2
 (4) 9.8 m/sec^2
22. The angle turned by a body undergoing circular motion depends on time as $\theta = \theta_0 + \theta_1 t + \theta_2 t^2$. Then the angular acceleration of the body is
- (1) θ_1 (2) θ_2
 (3) $2\theta_1$ (4) $2\theta_2$
23. A force acts on a 30 gm particle in such a way that the position of the particle as a function of time is given by $x = 3t - 4t^2 + t^3$, where x is in metres and t is in seconds. The work done during the first 4 seconds is
- (1) 5.28 J
 (2) 450 mJ
 (3) 490 mJ
 (4) 530 mJ
24. A body of mass 10 kg is dropped to the ground from a height of 10 metres. The work done by the gravitational force is ($g = 9.8\text{ m/sec}^2$)
- (1) - 490 Joules
 (2) + 490 Joules
 (3) - 980 Joules
 (4) + 980 Joules
25. A force $\vec{F} = 5\hat{i} + 6\hat{j} - 4\hat{k}$ acting on a body, produces a displacement $\vec{s} = 6\hat{i} + 5\hat{k}$. Work done by the force is
- (1) 18 units
 (2) 15 units
 (3) 12 units
 (4) 10 units
26. The work done against gravity in taking 10 kg mass at 1m height in 1sec will be
- (1) 49 J
 (2) 98 J
 (3) 196 J
 (4) None of these
27. A body of mass 6kg is under a force which causes displacement in it given by $S = \frac{t^2}{4}$ metres where t is time. The work done by the force in 2 seconds is
- (1) 12 J (2) 9 J
 (3) 6 J (4) 3 J
28. A body of mass 10 kg at rest is acted upon simultaneously by two forces 4 N and 3N at right angles to each other. The kinetic energy of the body at the end of 10 sec is
- (1) 100 J (2) 300 J
 (3) 50 J (4) 125 J
29. A force of $(3\hat{i} + 4\hat{j})$ Newton acts on a body and displaces it by $(3\hat{i} + 4\hat{j})\text{m}$. The work done by the force is
- (1) 10 J (2) 12 J
 (3) 16 J (4) 25 J
30. A force $\vec{F} = 6\hat{i} + 2\hat{j} - 3\hat{k}$ acts on a particle and produces a displacement of $\vec{s} = 2\hat{i} - 3\hat{j} + x\hat{k}$. If the work done is zero, the value of x is
- (1) -2 (2) 1/2
 (3) 6 (4) 2
31. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of force $4\hat{i} + \hat{j} + 3\hat{k}$ N. The work done will be
- (1) 100 J (2) 50 J
 (3) 200 J (4) 75 J
32. A body of mass 5 kg is placed at the origin, and can move only on the x-axis. A force of 10 N is acting on it in a direction making an angle of 60° with the x-axis and displaces it along the x-axis by 4 metres. The work done by the force is
- (1) 2.5 J (2) 7.25 J
 (3) 40 J (4) 20 J

33. A uniform chain of length 2m is kept on a table such that a length of 60cm hangs freely from the edge of the table. The total mass of the chain is 4kg. What is the work done in pulling the entire chain on the table
- (1) 7.2 J (2) 3.6 J
(3) 120 J (4) 1200 J
34. The potential energy of a certain spring when stretched through a distance 'S' is 10 joule. The amount of work (in joule) that must be done on this spring to stretch it through an additional distance 'S' will be
- (1) 30 (2) 40
(3) 10 (4) 20
35. Two springs of spring constants 1500 N/m and 3000 N/m respectively are stretched with the same force. They will have potential energy in the ratio
- (1) 4 : 1 (2) 1 : 4
(3) 2 : 1 (4) 1 : 2
36. A spring of spring constant 5×10^3 N/m is stretched initially by 5cm from the unstretched position. Then the work required to stretch it further by another 5cm is
- (1) 6.25 N-m (2) 12.50 N-m
(3) 18.75 N-m (4) 25.00 N-m
37. A mass of 0.5kg moving with a speed of 1.5 m/s on a horizontal smooth surface, collides with a nearly weightless spring of force constant $k = 50$ N/m. The maximum compression of the spring would be
- (1) 0.15 m (2) 0.12 m
(3) 1.5 m (4) 0.5 m
38. Which one of the following is not a conservative force
- (1) Gravitational force
(2) Electrostatic force between two charges
(3) Magnetic force between two magnetic dipoles
(4) Frictional force
39. Two bodies of masses m_1 and m_2 have equal kinetic energies. If p_1 and p_2 are their respective momentum, then ratio $p_1 : p_2$ is equal to
- (1) $m_1 : m_2$ (2) $m_2 : m_1$
(3) $\sqrt{m_1} : \sqrt{m_2}$ (4) $m_1^2 : m_2^2$
40. Two bodies of different masses m_1 and m_2 have equal momenta. Their kinetic energies E_1 and E_2 are in the ratio
- (1) $\sqrt{m_1} : \sqrt{m_2}$ (2) $m_1 : m_2$
(3) $m_2 : m_1$ (4) $m_1^2 : m_2^2$
41. Two bodies of masses 2m and m have their K.E. in the ratio 8 : 1, then their ratio of momenta is
- (1) 1 : 1 (2) 2 : 1
(3) 4 : 1 (4) 8 : 1
42. A 0.5 kg ball is thrown up with an initial speed 14 m/s and reaches a maximum height of 8.0m. How much energy is dissipated by air drag acting on the ball during the ascent
- (1) 19.6 Joule
(2) 4.9 Joule
(3) 10 Joule
(4) 9.8 Joule
43. A machine which is 75 percent efficient, uses 12 joules of energy in lifting up a 1 kg mass through a certain distance. The mass is then allowed to fall through that distance. The velocity at the end of its fall is (in ms^{-1})
- (1) $\sqrt{24}$ (2) $\sqrt{32}$
(3) $\sqrt{18}$ (4) $\sqrt{9}$
44. A force of $2\hat{i} + 3\hat{j} + 4\hat{k}$ N acts on a body for 4 second, produces a displacement of $(3\hat{i} + 4\hat{j} + 5\hat{k})$ m. The power used is
- (1) 9.5 W (2) 7.5 W
(3) 6.5 W (4) 4.5 W
45. A man does a given amount of work in 10 sec. Another man does the same amount of work in 20 sec. The ratio of the output power of first man to the second man is
- (1) 1 (2) 1/2
(3) 2/1 (4) None of these

46. A sealed container with gas at 2.00 atm is heated from 20.0 K to 40.0 K. The new pressure is
 (1) 0.50 atm (2) 1.00 atm
 (3) 4.00 atm (4) 2.14 atm
47. At the same temperature and pressure which of the following gas will have highest KE per mol?
 (1) H_2 (2) O_2
 (3) CH_4 (4) Equal
48. At what temperature will the total KE of 0.30 mole of He be the same as the total KE of 0.40 mole Ar at 400 K?
 (1) 533 K (2) 400 K
 (3) 346 K (4) 300 K
49. One molecule is moving at a speed of 1 m s^{-1} and two molecules are moving at 2 m s^{-1} . Hence, root mean square velocity is
 (1) 1 ms^{-1} (2) 1.67 ms^{-1}
 (3) 3 ms^{-1} (4) 1.73 ms^{-1}
50. The van der Waals' equation of state reduces itself to the ideal gas equation at
 (1) high pressure and low temperature
 (2) low pressure and low temperature
 (3) low pressure and high temperature
 (4) high pressure and high temperature
51. By what factor does the average velocity of a gaseous molecule increases, when the temperature (in Kelvin) is doubled?
 (1) 1.4 (2) 2.0
 (3) 2.8 (4) 4.0
52. A gaseous mixture was prepared by taking equal moles of CO and N_2 . If the total pressure of the mixture was found 1 atm, the partial pressure of the nitrogen (N_2) in the mixture is
 (1) 1 atm
 (2) 0.5 atm
 (3) 0.8 atm
 (4) 0.9 atm
53. The compressibility factor of gas is less than unity at STP, therefore
 (1) V_m (molar volume) $> 22.4 \text{ L}$
 (2) $V_m < 22.4 \text{ L}$
 (3) $V_m = 22.4 \text{ L}$
 (4) $V_m = 44.8 \text{ L}$
54. If V is the volume of one molecule of a gas under given conditions then van der Waals' constant b is
 (1) 4V
 (2) $\frac{4V}{N_0}$
 (3) $\frac{N_0}{4V}$
 (4) $4VN_0$
55. At what temperature will hydrogen molecules have the same KE per mol as nitrogen molecules at 280 K?
 (1) 280 K
 (2) 40 K
 (3) 400 K
 (4) 50 K
56. A 0.20 mole sample of a hydrocarbon C_xH_y after complete combustion with excess O_2 gas yields 0.80 mol of CO_2 and 1.0 mol of H_2O . Hence, hydrocarbon is
 (1) C_4H_{10}
 (2) C_4H_8
 (3) C_4H_5
 (4) C_8H_{16}
57. Which of the following is/are correct statement(s)?
 (1) van der Waals' constant 'a' is measure of attractive force
 (2) van der Waals' constant 'b' is also called co-volume or excluded volume
 (3) b is expressed in L mol^{-1}
 (4) All of the above

58. Assume that for every increase in height of 1 m, pressure decreases by 10 mmHg. Initially an experimental air balloon of maximum 200 L capacity has 150 L air at 1 atm at sea-level. At what height, the balloon is expected to burst?
- 15 m above sea-level
 - 19 m above sea-level
 - 10 m above sea-level
 - 20 m above sea-level
59. The density of phosphorus vapour at STP is 5.54 gL^{-1} . Hence, molecular formula of phosphorus is
- P_2
 - P_3
 - P_4
 - P_8
60. A 1 L flask contains 32 g O_2 gas at 27°C . What mass of O_2 must be released to reduce the pressure in the flask to 12.315 atm?
- 8 g
 - 16 g
 - 24 g
 - 0 g
61. 1 g H_2 and x g O_2 gas exert a total pressure of 5 atm. At a given temperature partial pressure of O_2 gas is 4 atm, O_2 in the mixture is
- 1 mol
 - 2 mol
 - 3 mol
 - 4 mol
62. 100 mL of H_2 gas diffuses in 10 s. X mL of O_2 gas diffuses in t s. X and t cannot be
- 100 mL, 40 s
 - 25 mL, 10 s
 - 100 mL, 10 s
 - 5 mL, 2 s
63. If $\bar{V}$ is the observed molar volume of real gas and $\bar{V}_{\text{id}}$ is the molar volume of an ideal gas then Z is
- $\bar{V}\bar{V}_{\text{id}}$
 - $\frac{\bar{V}}{\bar{V}_{\text{id}}}$
 - $\frac{\bar{V}_{\text{id}}}{\bar{V}}$
 - $\frac{\bar{V}^2}{\bar{V}_{\text{id}}}$
64. Densities of CO and N_2 are in the ratio of 1 : 2 and their temperatures are in the ratio of 2 : 1, thus the ratio of their pressure is
- 1 : 1
 - 1 : 2
 - 2 : 1
 - 4 : 1
65. If the four tubes of a car are filled to the same pressure with N_2 , O_2 , H_2 and Ne separately, then which one will be filled first?
- N_2
 - O_2
 - Ne
 - H_2
66. For the following gaseous equilibria X, Y and Z at 300 K
- $$\text{X} : 2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$$
- $$\text{Y} : \text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$$
- $$\text{Z} : 2\text{HI} \rightleftharpoons \text{H}_2 + \text{I}_2$$
- ratio of K_p and K_c in the increasing order is
- $X = Y = Z$
 - $X < Y < Z$
 - $X < Z < Y$
 - $Z < Y < X$
67. For the equilibrium,
- $$\text{I} : \text{C}_6\text{H}_5\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{C}_6\text{H}_5\text{COO}^- + \text{H}_3\text{O}^+;$$
- $$K_1 = 6.30 \times 10^{-5}$$
- $$\text{II} : \text{C}_6\text{H}_5\text{COOH} + \text{OH}^- \rightleftharpoons \text{C}_6\text{H}_5\text{COO}^- + \text{H}_2\text{O};$$
- $$K_2 = 6.30 \times 10^9$$
- $$\text{III} : \text{H}_2\text{O} + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{OH}^-; K_3 = ?$$
- K_3 using the above equilibria is
- $(6.30)^2 \times 10^4$
 - 1.0×10^{-14}
 - 1×10^{14}
 - $(6.30)^{-2} \times 10^{-4}$
68. For the equilibrium,
- $$\text{NH}_2\text{COONH}_4(\text{s}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g})$$
- $P_{\text{CO}_2} = 1 \text{ atm}$ at 100°C . Hence, equilibrium constant is
- 1 atm³
 - 2 atm³
 - 4 atm³
 - 3 atm³

69. For the following equilibrium reaction,



NO_2 is 50% of the total volume at a given temperature. Hence, vapour density of the equilibrium mixture is

- (1) 34.5 (2) 25.0
(3) 23.0 (4) 20.0

70. For a very small extent of dissociation of PCl_5 into PCl_3 and Cl_2 is a gaseous phase reaction then degree of dissociation x

- (1) $x \propto p$
(2) $x \propto \frac{1}{p}$
(3) $x \propto \sqrt{p}$
(4) $x \propto \sqrt{\frac{1}{p}}$

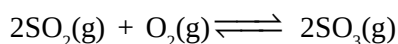
71. 60 g CH_3COOH and 46 g $\text{CH}_3\text{CH}_2\text{OH}$ react in 5L flask to form 44 g $\text{CH}_3\text{COOC}_2\text{H}_5$ at equilibrium. On taking 120 g CH_3COOH and 46 g $\text{CH}_3\text{CH}_2\text{OH}$, $\text{CH}_3\text{COOC}_2\text{H}_5$ formed at equilibrium is

- (1) 44 g (2) 20.33 g
(3) 22 g (4) 58.66 g

72. If for $\text{A} \rightleftharpoons \text{B}$, $K = 16$ then K for $\frac{1}{2}\text{B} \rightleftharpoons \frac{1}{2}\text{A}$ will be :-

- (1) 4 (2) $\frac{1}{4}$
(3) 2 (4) $\frac{1}{2}$

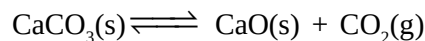
73. In the manufacturing of H_2SO_4 with the following equilibrium



reaction proceeds in the forward direction if

- (1) some H_2O is added
(2) some SO_3 is added
(3) some SO_2 is removed
(4) some O_2 is removed

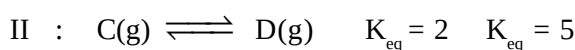
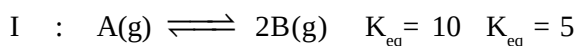
74. Following equilibrium is disturbed by



- (1) addition of CaCO_3
(2) addition of CaO
(3) addition of CO_2
(4) All of these

75. Consider the following equilibria at 300 K and 400 K with their equilibrium constants

300 K 400 K



Thus,

- (1) I is endothermic, II is exothermic
(2) I is exothermic, II is endothermic
(3) I and II both are endothermic
(4) I and II both are exothermic

76. Strongest acid and strongest base pair is

- (1) H_3O^+ , H_2O
(2) H_3O^+ , OH^-
(3) H_2O , H_3O^+
(4) OH^- , H_3O^+

77. pH of 10^{-8} N NaOH is

- (1) 8.0 (2) 6.0
(3) 6.98 (4) 7.02

78. Milliequivalent of $\text{Ba}(\text{OH})_2$ present in 100 mL solution to have pH = 13, is

- (1) 10^{-2} (2) 10
(3) 0.1 (4) 0.01

79. A weak monobasic acid is 0.1% ionised at 0.1 M. Hence, its pH is

- (1) 2 (2) 3
(3) 4 (4) 5

80. Which has maximum pH?

- (1) 0.01 M H_2SO_4
(2) 0.01 M HCl
(3) 0.01 M $\text{Ca}(\text{OH})_2$
(4) 0.01 M NaOH

81. 50.0 mL of 0.3 M HCl is mixed with 50 mL of 0.4 M NH_3 solution. If pK_a of NH_4^+ is 9.26, pH of the mixture is

- (1) 5.22
- (2) 1.30
- (3) 8.78
- (4) 12.70

82. $[\text{Ag}^+]$ in saturated AgCl in presence of 1 M KCl [$K_{sp}(\text{AgCl}) = 1 \times 10^{-10}$] is

- (1) 1×10^{-5} M
- (2) 1×10^{-20} M
- (3) 1×10^{-11} M
- (4) 1×10^{-10} M

83. K_{sp} of $\text{Al}(\text{OH})_3$ is 1.0×10^{-15} . pH of the saturated solution is about

- (1) 5.0
- (2) 9.0
- (3) 4.1
- (4) 10.4

84. pH of 0.01 M aqueous solution of NaX, NaY and NaZ are 8, 9 and 10 respectively. Strongest acid of HX, HY and HZ is

- (1) HX
- (2) HY
- (3) HZ
- (4) Can't be predicted

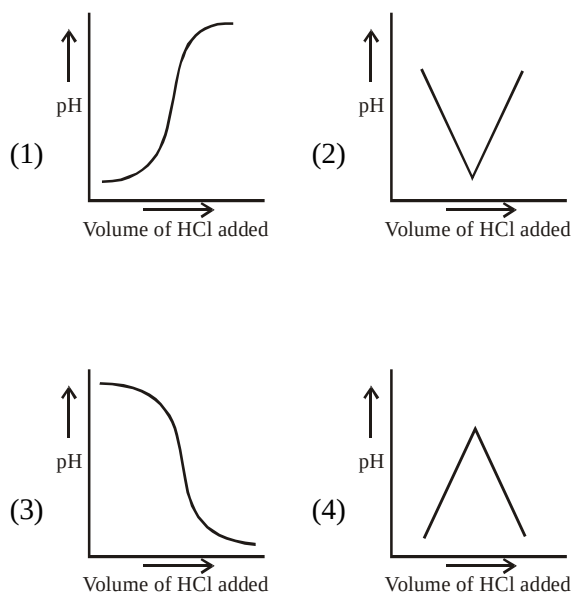
85. pH of 0.05 M calcium acetate solution ($\text{pK}_a = 4.74$) is

- (1) 8.72
- (2) 8.87
- (3) 7.00
- (4) 1.30

86. pK_{a_1} , pK_{a_2} and pK_{a_3} of H_3PO_4 are respectively x, y and z. pH of 0.01 M Na_2HPO_4 solution is

- (1) 2
- (2) $\left(\frac{x+y}{2}\right)$
- (3) $\left(\frac{y+z}{2}\right)$
- (4) $\left(\frac{x+y+z}{2}\right)$

87. Titration curve if a strong base is titrated with strong acid is



88. 100 mL of 0.1 M HCl is titrated using 0.1 M NaOH, using phenolphthalein indicator. pH after 50 mL of NaOH has been added is

- (1) 7.0
- (2) 3.48
- (3) 6.0
- (4) 1.48

89. At 4°C , $K_w = 1 \times 10^{-16}$. A solution with $\text{pH} = 7.5$ at 4°C will

- (1) turn blue litmus red
- (2) turn red litmus blue
- (3) turn turmeric paper brown
- (4) be neutral to litmus

90. The solubility of A_2X_3 is S mol L^{-1} . Its solubility product is

- (1) $6S^4$
- (2) $64S^4$
- (3) $36S^5$
- (4) $108S^5$

91. On the basis of given diagram, identify the incorrect statement :-



- (1) 10 stamens are arrangement in two groups
(2) Carpel is single
(3) Polysepalous with imbricate aestivation
(4) Polypetalous with vexillary aestivation
92. Trimerous flower, superior ovary with axile placentation are characteristic of
(1) Liliaceae (2) Cucurbitaceae
(3) Solanaceae (4) Asteraceae
93. Brinjal, Potato, Cabbage, Onion, Mustard belong to
(1) Four family (2) Four genera
(3) Three genera (4) Four species
94. Match the columns I and II, and choose the correct combination from the options given.

Column-I		Column-II	
(a)		1.	Brassicaceae
(b)		2.	Fabaceae
(c)		3.	Solanaceae
(d)		4.	Liliaceae

- (1) a-3, b-2, c-1, d-4
(2) a-2, b-4, c-1, d-3
(3) a-1, b-3, c-4, d-2
(4) a-2, b-4, c-3, d-1

95. Diadelphous condition occurs in

- (1) Solanaceae
(2) Fabaceae
(3) Asteraceae
(4) Liliaceae

96. Match the columns I and II, and choose the correct combination from the options given.

Column I (Medicinal plant)		Column II (Family)	
a.	<i>Aloe</i>	1.	Brassicaceae
b.	<i>Belladonna</i>	2.	Fabaceae
c.	<i>Muliathi</i>	3.	Solanaceae
d.	<i>Ashwagandha</i>	4.	Liliaceae

- (1) a-1, b-2, c-3, d-4
(2) a-4, b-3, c-2, d-1
(3) a-4, b-3, c-2, d-3
(4) a-4, b-2, c-3, d-3

97. An aspect of flower shown in floral formula but not in floral diagram is

- (1) Position of ovary
(2) Floral symmetry
(3) Aestivation
(4) Cohesion of floral parts

98. Match the columns I, II and III and choose the correct combination from the options given.

Column I		Column II		Column III	
a.	$\text{Br} \oplus \overset{\curvearrowright}{\text{P}}_{(3+3)} \text{A}_{(3+3)} \underline{\text{G}}_{(3)}$	1.	Solanaceae	K.	Mustard
b.	$\% \oplus \overset{\curvearrowright}{\text{K}}_{(5)} \text{C}_{1+2+2} \text{A}_{(9)+1} \underline{\text{G}}_1$	2.	Liliaceae	L.	<i>Indigofera</i>
c.	$\oplus \overset{\curvearrowright}{\text{K}}_{(5)} \text{C}_{(5)} \text{A}_2 \underline{\text{G}}_{(2)}$	3.	Brassicaceae	M.	<i>Petunia</i>
d.	$\oplus \overset{\curvearrowright}{\text{K}}_{2+2} \text{C}_4 \text{A}_{2+4} \underline{\text{G}}_{(2)}$	4.	Fabaceae	N.	<i>Tulip</i>

- (1) a-2-M, b-1-L, c-3-K, d-4-N
(2) a-2-N, b-4-L, c-1-M, d-3-K
(3) a-2-M, b-4-N, c-1-L, d-3-K
(4) a-2-N, b-4-M, c-3-K, d-1-L

99. Which set of angiosperms belong to same angiosperm family?

- (1) Chilli, Petunia, Aloe
- (2) Aloe, Gloriosa, Colchicum
- (3) Lupin, Indigofera, Potato
- (4) Soyabean, Moong Brinjal

100. Observe the given floral diagram. Select the option with suitable example in which it is found :-



- (1) Pea, Gram
- (2) Tomato, Brinjal
- (3) Sunflower, Bean
- (4) Onion, Tulip

101. Floral diagram fails to indicate

- (1) epiphyllly and epipetaly
- (2) aestivation and placentation
- (3) position of ovary on the thalamus
- (4) cohesion of carpels and stamens

102. $\text{♂K}_{(5)}\text{C}_{(5)}\text{A}_{5}\text{G}_{(2)}$ is the floral formula of

- (1) *Petunia*
- (2) *Brassica*
- (3) *Allium*
- (4) *Sesbania*

103. The distinct features of fabaceae are

- (1) Zygomorphic, diadelphous and monocarpellary
- (2) Actinomorphic, monadelphous and monocarpellary
- (3) Zygomorphic, monadelphous and pentacarpellary
- (4) Zygomorphic, polyadelphous and tricarpeal

104. Colchicine is obtained from *Colchicum autumnale*. It belongs to family

- (1) Liliaceae
- (2) Solanaceae
- (3) Leguminaceae
- (4) Poaceae

105. Pentamerous actinomorphic flowers, bicarpellary ovary with oblique septum and a fruit of capsule or berry, belong to family

- (1) Liliaceae
- (2) Asteraceae
- (3) Brassicaceae
- (4) Solanaceae

106. Which of the following feature is not associated with pea?

- (1) Marginal placentation
- (2) Legume fruit
- (3) Papilionaceous corolla
- (4) Monocarpellary inferior ovary

107. A family having perennial herbs with underground structures as bulbs, corms rhizomes and rarely berry fruit type is identified as

- (1) Liliaceae
- (2) Solanaceae
- (3) Fabaceae
- (4) Malvaceae

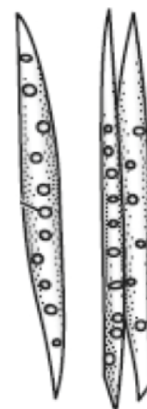
108. *Indigofera*, *sesbania* and *trifolium* are the members of family

- (1) Fabaceae
- (2) Asteraceae
- (3) Solanaceae
- (4) Liliaceae

109. Diagnostic feature of angiospermic family that includes belladonna and ashwagandha

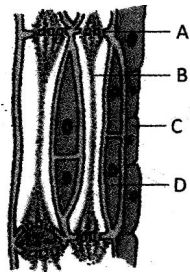
- (1) Didelphous stamens
- (2) Marginal placentation
- (3) Bicarpellary syncarpous superior ovary
- (4) Underground bulb, corm rhizome

110. Identify the structure given and select the option with incorrect information regarding it :-



- (1) Given cells are dead in nature
- (2) Walls are heavily suberised
- (3) Component of xylem
- (4) Provide mechanical support to the hadrome

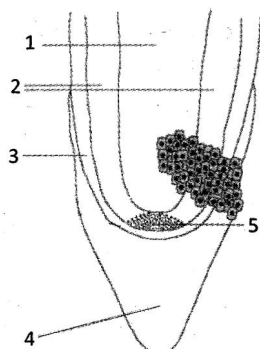
111. Which of the following is not correctly matched with its feature?
- (1) Trichomes—Multicellular
 - (2) Guard cells in dicots—Bean-shaped
 - (3) Hemp fibre—Bast fibre
 - (4) Cork cambium—Secondary growth in stellar region
112. Typical dicot stem possess open vascular bundles. What is incorrect about such vascular bundles?
- (1) Inter-fascicular cambium is present between xylem and phloem
 - (2) It may show secondary growth
 - (3) It may give rise to secondary xylem and secondary phloem both
 - (4) Cambium of such bundles is example of secondary meristem
113. Diagram of phloem is given in which A-D are labelled. Which labelling is not correctly indicating its feature/function?



- (1) A-Facilitate transport through sieve tube elements
 - (2) C-Cell wall is lignified
 - (3) D-Maintain pressure gradients in sieve tubes
 - (4) B-Possess peripheral cytoplasm but no nucleus
114. Starch sheath is found in :-
- (1) Monocot leaf
 - (2) Dicot leaf
 - (3) Monocot root
 - (4) Dicot stem

115. If xylem and phloem are arranged at different radii, then arrangement of vascular bundles is said to be :-
- (1) Open
 - (2) Closed
 - (3) Radial
 - (4) Endarch
116. How many of the following are examples of lateral meristem?
- [Cork cambium, Intercalary meristem, Intrafascicular cambium, Interfascicular cambium, Apical meristem]**
- (1) 4
 - (2) 3
 - (3) 2
 - (4) 1
117. Parenchymatous cells found between xylem and phloem in dicot root are termed as :-
- (1) Casparian strips
 - (2) Conjunctive tissue
 - (3) Starch sheath
 - (4) Stele
118. Read the statements (A-E) and answer the question following them.
- (A) Permanent tissues in plants do not have the capability of division.
 - (B) After grazing by herbivores, grasses regenerate their lost parts by the action of intercalary meristem.
 - (C) Interfascicular cambium is an example of apical meristem.
 - (D) Metaxylem is a type of primary xylem.
 - (E) Vessels are found in almost all seed plants.
- How many statements are false?
- (1) 1
 - (2) 2
 - (3) 3
 - (4) 4

119. Section of root is given. Select the option with correct identification:



- (1) 1-pith, 3-intercalary meristem
(2) 2-cortex, 5-root apical meristem
(3) 1-central cylinder, 5-lateral meristem
(4) 3-protoderm, 4-root apical meristem

120. Which meristem is not correctly matched with its location and function?

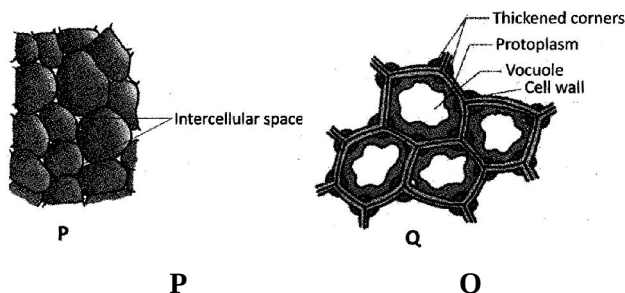
(1)	Apical meristem	Shoot and root apex	Elongation of stem and root
(2)	Intercalary meristem	In between the apical meristem	Growth in grasses
(3)	Intrafascicular cambium	In the vascular bundles	Increases the girth of stem / root
(4)	Interfascicular cambium	In between vascular bundles	Increases the girth of stem / root

121. Fill in the blanks :-

Those permanent tissues which have all similar cells are termed as ____ 1 ____ tissues, in plants ____ 1 ____ tissues are of 3 types ____ 2 ____, ____ 3 ____ and ____ 4 ____ Cells of ____ 2 ____ are living cells without intercellular spaces. These provide mechanical support. Cells of ____ 3 ____ are generally isodiametric and form the major component of plants organs. Cells of ____ 4 ____ possess lignified cell walls and are usually dead.

- (1) 1 = simple, 2 = parenchyma, 3 = collenchyma, 4 = sclerenchyma
(2) 1 = simple, 2 = collenchyma, 3 = parenchyma, 4 = sclerenchyma
(3) 1 = complex, 2 = collenchyma, 3 = sclerenchyma, 4 = parenchyma
(4) 1 = simple, 2 = parenchyma, 3 = collenchyma, 4 = sclerenchyma

122. P and Q are two types of simple permanent tissues. Select the correct option with their occurrence :-



- (1) Cortex of dicot stem Below epidermis in dicot stem
(2) Pulp of guava Pith in monocot root
(3) Medullary rays Epidermis of stem
(4) Xylem and phloem Lignified fibres

123. Meristematic cells do not have

- (1) Metabolic activities
(2) Large vacuoles
(3) Ergastic substances
(4) More than one option is correct

124. During the formation of primary plant body, specific regions of the apical meristem produces

- (1) Epidermis
(2) Cortex
(3) Vascular tissue
(4) All of these

125. Parenchyma are found almost in all parts of plants and can be characterized by

- (1) Presence of thin cellulose cell wall
(2) Non-vacuolated cytoplasm
(3) Thick pectocellulosic cell wall
(4) Presence of pits in its cell wall

126. Axillary bud is formed by the cells which are left behind from

- (1) Lateral meristem
(2) Intercalary meristem
(3) Shoot apical meristem
(4) Root apical meristem

127. Which of these is responsible for grittiness of fruit?

- (1) Stone cells
- (2) Rod cells
- (3) Star cells
- (4) Collenchymatous tissue

128. Lateral meristem occurs in the mature regions of

- (1) Shoot and leaves of monocot plants
- (2) Roots and shoots of monocot plants
- (3) Roots and shoots of dicot plants
- (4) Roots and leaves of dicot plants

129. Select the **incorrect** statement w.r.t. phloem.

- (1) Complex permanent tissue which transport food materials
- (2) In gymnosperm phloem lack sieve tubes and companion cells
- (3) Protophloem has bigger sieve tubes
- (4) Phloem fibres are absent in primary phloem

130. Lateral roots help in efficient absorption and are endogenous in origin, they arise from

- (1) Pericycle
- (2) Endodermis
- (3) Cortex
- (4) Pith

131. Which of the following statement is **correct** w.r.t. development of primary xylem?

- (1) Exarch condition in monocot stem
- (2) Endarch condition in dicot root
- (3) Centripetal formation of xylem in dicot root
- (4) Centrifugal formation of xylem in monocot root

132. The rolling of monocot leaves in summer is due to loss of water from

- (1) Motor cells
- (2) Endodermal cells
- (3) Cortex
- (4) None of these

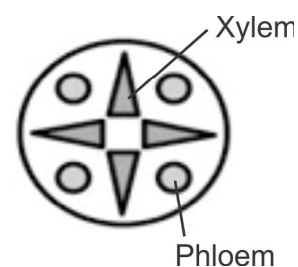
133. Read the following statement carefully:

- (a) Present below epidermis in monocot stem
- (b) Cells are dead without protoplast with lignified cell wall
- (c) Provide mechanical strength

These all are features of

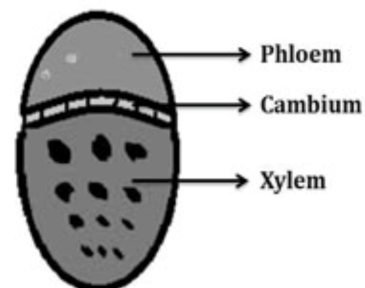
- (1) Collenchyma
- (2) Sclerenchyma
- (3) Aerenchyma
- (4) Prosenchyma

134. The given vascular bundle is present in :-



- (1) Monocot root
- (2) Dicot root
- (3) Dicot stem
- (4) Isobilateral leaf

135. The following vascular bundles are present in :-



- (1) Mustard root
- (2) Sunflower stem
- (3) Sunflower leaf
- (4) Mango root

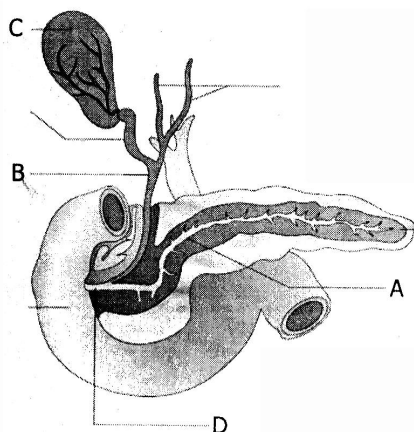
136. Which of the following layer of alimentary canal is highly vascular in nature :

- (1) Serosa
- (2) Muscularis
- (3) Submucosa
- (4) Mucosa

137. Select the duct of gall bladder :

- (1) Haptic duct
- (2) Stenson's duct
- (3) Common hepato-pancreatic duct
- (4) Cystic duct

138. Identify A, B, C and D in the given diagram :



- (1) A = Pancreatic duct, B = Common bile duct, C = Gall bladder, D = Hepato-pancreatic duct
- (2) A = Common bile duct, B = Pancreatic duct, C = Gall bladder, D = Hepato-pancreatic duct
- (3) A = Pancreatic duct, B = Common bile duct, C = Hepato-pancreatic duct, D = Gall bladder
- (4) A = Hepato-pancreatic duct, B = Common bile duct, C = Gall bladder, D = Pancreatic duct

139. The chemical process of digestion is initiated by the action of :

- (1) Salivary Amylase
- (2) Maltase
- (3) Peptidase
- (4) Trypsin

140. The stomach stores the food for :

- (1) 1 - 2 hrs.
- (2) 2 - 3 hrs.
- (3) 3 - 4 hrs.
- (4) 4 - 5 hrs.

141. Read the statements :

- (i) The common hepatopancreatic duct is guarded by sphincter of oddi.
- (ii) The endocrine portion of pancreas secretes an alkaline pancreatic juice.
- (iii) The intrinsic factor (factor essential for absorption of Vitamin B₁₂) is secreted by oxyntic cells.
- (iv) Sub - mucosal glands are also known as Brunner's glands.

How many statements of the above are true :

- (1) 4
- (2) 3
- (3) 2
- (4) 1

142. Which one of the following is not a cell of gastric glands :

- (1) Mucus neck cells
- (2) Paneth cells
- (3) Peptic cells
- (4) Parietal cells

143. Which component of food is hydrolysed in oral cavity by the action of salivary amylase ?

- (1) fat
- (2) carbohydrate
- (3) protein
- (4) all of these

144. If for some reason the parietal cells of the gut epithelium become partially non-functional, what is likely to happen

- (1) The pancreatic enzymes specially the trypsin and lipase will not work efficiently.
- (2) The pH of stomach will fall.
- (3) The secretion of pepsinogen will be decreased.
- (4) Proteins will not be adequately hydrolysed by pepsin into proteoses and peptones.

145. Match the Column I with Column II and choose the correct option :

Column - I

Column - II

- | | |
|------------------------------|---------------|
| (a) Bilirubin and Biliverdin | (i) Parotid |
| (b) Hydrolysis of starch | (ii) Bile |
| (c) Digestion of fat | (iii) Lipases |
| (d) Salivary gland | (iv) Amylases |

- (1) a = (ii), b = (iv), c = (iii), d = (i)
 (2) a = (i), b = (ii), c = (iii), d = (iv)
 (3) a = (iv), b = (iii), c = (ii), d = (i)
 (4) a = (i), b = (iv), c = (iii), d = (ii)

146. How many human teeth appear twice during the life span of an individual :

- (1) 12 (2) 20
 (3) 28 (4) 32

147. Jaundice may be caused by retarded function of :

- (1) Stomach
 (2) Liver
 (3) Intestines
 (4) Pancreas

148. Match the Column I with Column II and choose the correct option :

Column I

Column II

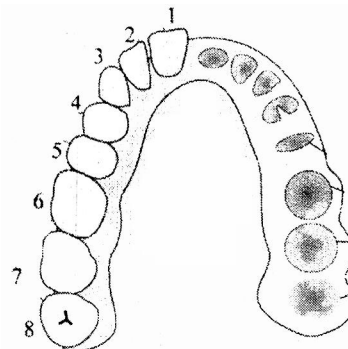
- | | |
|------------------|------------------------|
| A. Goblet cells | 1. Antibacterial agent |
| B. Lysozymes | 2. Mucus |
| C. Saliva | 3. HCl |
| D. Oxyntic cells | 4. Sublingual gland |

- (1) A = 3, B = 1, C = 4, D = 2
 (2) A = 1, B = 3, C = 4, D = 2
 (3) A = 2, B = 3, C = 1, D = 4
 (4) A = 2, B = 1, C = 4, D = 3

149. Regarding the digestive system which is true :

- (1) The bases of villi in the intestine have crypts of langerhans.
 (2) The cystic duct directly opens into duodenum and carries bile.
 (3) The bile helps in emulsification and also helps in activity of lipase.
 (4) The exocrine part of pancreas secretes the acidic pancreatic juice.

150. The diagram of human jaw is given showing the teeth. The names of teeth numbered 3 and 4 are :-



- (1) Incisor and Canine
 (2) Canine and Premolar
 (3) Premolar and molar
 (4) Incisor and Premolar

151. Which of the following is not a disorder of digestive system :

- (1) Jaundice
 (2) Vomiting
 (3) Angina pectoris
 (4) Constipation

152. Match the following :

Column I

Column II (pH)

- | | |
|-----------------------|---------|
| I. Acid in stomach | (a) 6.8 |
| II. Salivary amylase | (b) 7.8 |
| III. Pancreatic juice | (c) 1.8 |

- (1) I = a, II = b, III = c
 (2) I = b, II = a, III = c
 (3) I = c, II = a, III = b
 (4) I = c, II = b, III = a

153. Bile is formed in :

- (1) Gall bladder
 (2) Liver
 (3) Spleen
 (4) Blood

154. Proteolytic enzymes do not corrode lining of alimentary canal as :

- (1) They are secreted in inactive form
 (2) Lining layer of alimentary canal does not contain protein
 (3) The enzymes are not capable of digesting fat
 (4) None of these

155. In stomach which one comes :

- (1) Bolus
- (2) Chyme
- (3) Chyle
- (4) All

156. Which one correctly explains the condition of constipation :

- (1) The food is not properly digested leading to a feeling of fullness.
- (2) The faeces are retained within the rectum as the bowel movements occur irregularly.
- (3) It is the ejection of stomach contents through the mouth.
- (4) The abnormal frequency of bowel movement and increased liquidity of the faecal discharge.

157. Where do certain symbiotic microorganism normally occurs in human body ?

- (1) Vermiform appendix and rectum
- (2) Duodenum
- (3) Caecum
- (4) Oral lining and tongue surface

158. Emulsification of fat will not occur in the absence of :

- (1) Lipase
- (2) Bile pigments
- (3) Bile salts
- (4) Pancreatic juice

159. The distal part of the stomach that opens into duodenum is called :

- (1) Fundus
- (2) Pylorus
- (3) Omentum
- (4) Cardiac

160. Which of the following is the wrong match in the given table?

Options :

		Parotid gland	Submaxillary gland	Sublingual gland
1	Location	Below ear	Jaw angle	Below tongue
2	Number	One pair	One pair	One pair
3	Duct	Stenson duct	Duct of Rivinus	Wharton duct
4	Secretion	Saliva + enzyme	Saliva	Saliva

161. The structural and functional unit of liver is :

- (1) Glisson's capsule
- (2) Hepatic lobules
- (3) Hepatic vein
- (4) Hepatic cords

162. Which part of alimentary canal does not secrete any digestive enzyme ?

- (1) Mouth
- (2) Stomach
- (3) Duodenum
- (4) Oesophagus

163. The epithelial cells lining the stomach of vertebrates is protected from the damage by HCl because :

- (1) HCl is too dilute
- (2) HCl is neutralized by stomach
- (3) The epithelial cells are resistant to the action of HCl
- (4) The epithelial cells are covered by a mucous secretion.

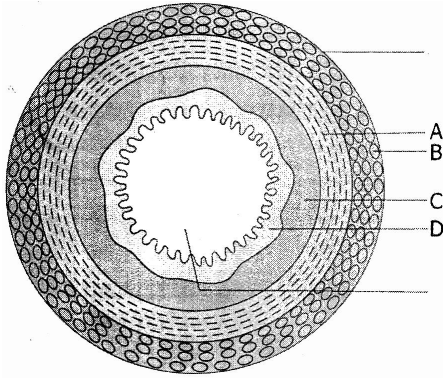
164. Which of the following correctly shows the composition of saliva ?

- (1) Electrolytes, Salivary amylase and Lipase
- (2) Electrolytes, Dipeptidase and Lysozyme
- (3) Electrolytes, Salivary amylase and Lysozyme
- (4) Salivary amylase, Lipase and Lysozyme

165. Pulp cavity of teeth is surrounded by :

- (1) Enamel
- (2) Gums
- (3) Bones
- (4) Odontoblasts

166. Identify A, B, C and D in the given diagram:



- (1) A = Outer longitudinal muscle, B = Inner circular muscle, C = Sub - mucosa, D = Mucosa
- (2) A = Inner circular muscle, B = Outer longitudinal muscle, C = Sub- mucosa, D = Mucosa
- (3) A = Inner circular muscle, B = Outer longitudinal muscle, C = Mucosa, D = Sub-mucosa
- (4) A = Sub- mucosa, B = Mucosa, C = Outer longitudinal muscle, D = Inner circular muscle

167. Which one of the following pairs of kind of cells and their function is not correctly matched?

- (1) Parietal cells - secrete hydrochloric acid and intrinsic factor
- (2) Ameloblasts - secrete dentine
- (3) Zymogen cells- secrete pepsinogen
- (4) Paneth cells - secrete enzymes

168. During prolonged fasting, in what sequence are the following organic compounds consumed by the body :-

- (1) First carbohydrates, next fats and lastly proteins
- (2) First fats, next carbohydrates and lastly proteins
- (3) First carbohydrates, next proteins and lastly lipids.
- (4) First proteins, next lipids and lastly carbohydrates.

169. Which one of these are most essential for the body growth and formation of new cells ?

- (1) Fats
- (2) Sugars
- (3) Vitamins
- (4) Proteins

170. Herbivorous animals can digest cellulose because-

- (1) they have large alimentary canal
- (2) Bacteria present in their caecum help in the digestion of cellulose.
- (3) Gastric juice has a digestive enzyme for cellulose digestion
- (4) their molar and premolar teeth can crush and grind the food

171 Peyer's patches are found in

- (1) Lamina propria of ileum
- (2) submucosa of duodenum
- (3) Serosa of jejunum
- (4) Muscularis of stomach

172. The stomach doesnot digest itself because

- (1) Gastric mucosa is protected by thick layer of mucus
- (2) Gastric mucosal cells are not digestible
- (3) Gastric mucosal cells transport hydrogen ions out of the gastric mucosa
- (4) Hydrogen ions are completely neutralized by mucus

173. Find the odd one out

- (1) duodenum
- (2) jejunum
- (3) ileum
- (4) caecum

174. _____ covers the opening of the internal nasal pores during ingestion of food , so food particle cannot move inside nasal chamber.

- (1) uvula
- (2) palate
- (3) tonsils
- (4) buccal vestibule

175. Which of the following statement is **incorrect** about enamel ?

- (1) ectodermal in origin
- (2) secreted by cementocytes
- (3) hardest substance of animal kingdom
- (4) not present in sloth

176. Match the following columns and choose the correct answer from the options given below

	Column A		Column B
A	Incisors	i	Gnawing of food
B	Canines	ii	Sharp , pointed
C	Premolars	iii	Chewing of food
D	Molars		

- | A | B | C | D |
|---------|-----|-----|-----|
| (1) i | ii | iii | iii |
| (2) i | iii | ii | iii |
| (3) i | ii | i | iii |
| (4) iii | i | ii | iii |

177. Which of the following is not the characteristic feature of human teeth?

- (1) thecodont
- (2) diphyodont
- (3) heterodont
- (4) polyphyodont

178. Three major type of cells in stomach are

- (1) mucous cells , chief cells , oxyntic cells
- (2) mucous cells , chief cells , enteroendocrine cells
- (3) mucous cells , enteroendocrine cells , oxyntic cells
- (4) enteroendocrine cells ,chief cells , oxyntic cells

179. Which of the following statement is correct?

- (1) duodenum is the longest part of small intestine
- (2) caecum continues into colon which is the initial part of the large intestine
- (3) rectum opens into anal canal
- (4) two types of sphincter muscles are found at the opening of anus – internal which is voluntary and external which is involuntary

180. Which of the following is / are modification /s of histology of alimentary canal ?

- (a) multilayered epithelium in oesophagus
 - (b) oblique muscle layers in stomach
 - (c) Brunners glands in submucosa of duodenum
 - (d) Paneth cells in mucosa of the jejunum
- (1) a and b only (2) b and c only
(3) d only (4) a , b , c and d all



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